Reproduction: Reverse Recovery in Silicon Power Rectifiers

H. Benda and E. Spenke, Proc. IEEE 55(8), 1331-1354, Aug. 1967

High-injection p-s-n / p-i-n rectifier. The excess plasma obeys
the ambipolar diffusion equation dn/dt = D n_xx - n/tau with
high-level current boundary conditions. Two independent solvers:
  - closed form Eq. (21) forward, Eq. (36) transient
  - numerical implicit ambipolar PDE (Neumann BCs)

Reproduced figures:
  Fig. 4 stationary forward profile (asymmetric catenary)
  Fig. 7 decay by recombination (iR = 0)
  Fig. 8 sweeping-out by strong reverse current (iR = 10 iF)
  Fig. 9 storage times t0l, t0r vs iR/iF (d=L and d=L/2)

See README.md and derivation.md for the full derivation.

Model vs paper -- verification

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Benda & Spenke 1967 -- reverse recovery -- model vs paper / self-checks
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[ material / device (silicon, mu_n = 3 mu_p) ]
  B = (mu_n-mu_p)/(mu_n+mu_p) = 0.5000   (paper: B = 0.5)
  D (ambipolar)  = 17.450 cm^2/s
  L (ambipolar)  = 132.10 um
  nbar           = 2.362e+17 cm^-3   (from iF = 2 d q nbar/tau)

[ exact identities (should be exact) ]
  <n_forward> / nbar                = 1.000000   (= 1, Eq. 24)
  Eq.36(t=0) vs Eq.21 forward, RMS   = 2.62e-04 of nbar

[ pure recombination decay, iR=0: border-avg vs exp(-t/tau) ]
  t/tau=1 : <n>/nbar = 0.3679   exp(-t/tau) = 0.3679
  t/tau=2 : <n>/nbar = 0.1353   exp(-t/tau) = 0.1353
  t/tau=3 : <n>/nbar = 0.0498   exp(-t/tau) = 0.0498

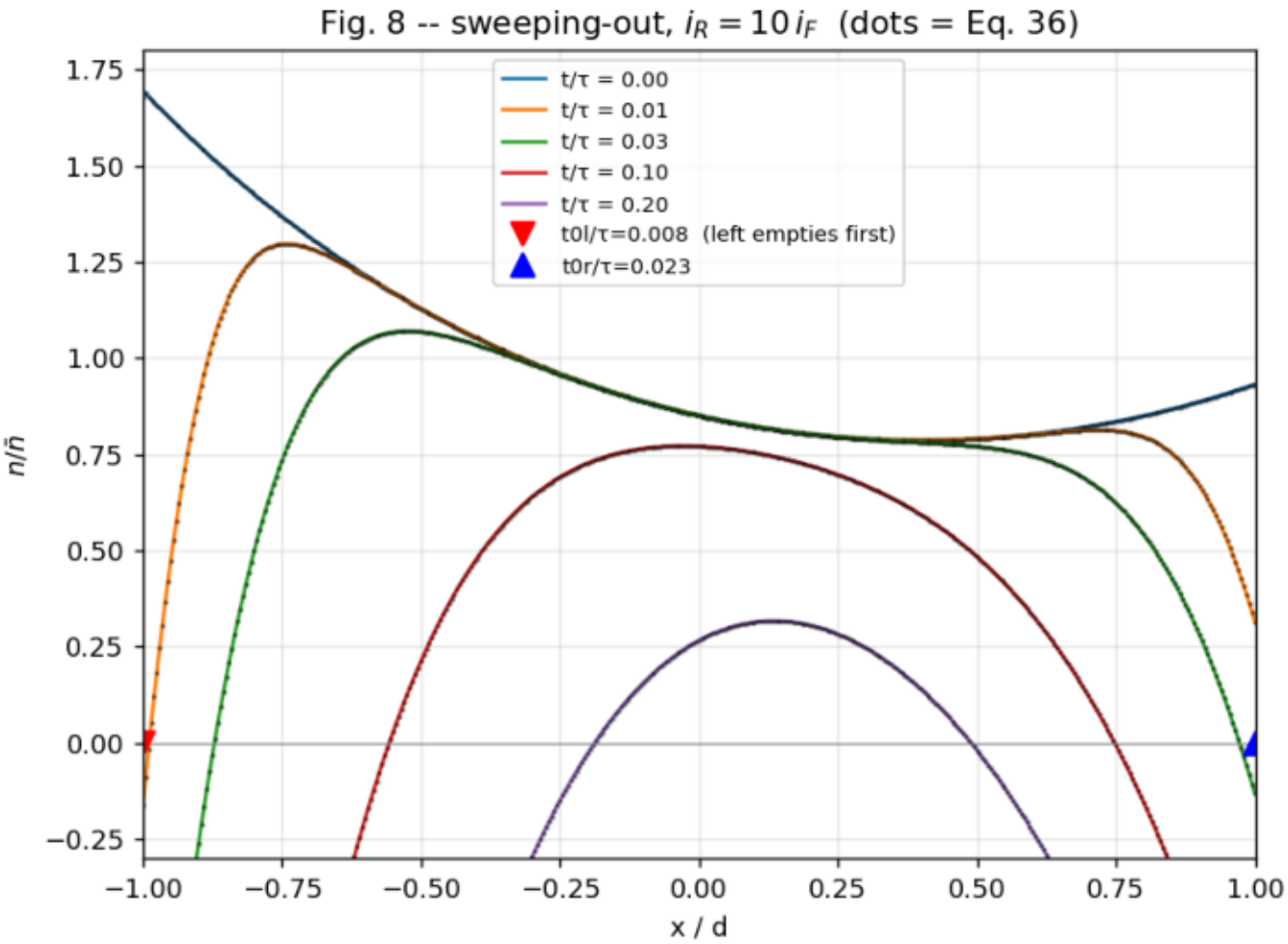
[ storage times: analytic (Eq. 36) vs numerical PDE ]
  d/L  iR/iF |  t0l/tau (an / num)  |  t0r/tau (an / num)
  1.0     2 |   0.1204 /  0.1205   |   0.3535 /  0.3535
  1.0    10 |   0.0083 /  0.0084   |   0.0229 /  0.0229
  1.0    50 |   0.0004 /  0.0004   |   0.0011 /  0.0011
  0.5     2 |   0.2343 /  0.2343   |   0.4384 /  0.4384
  0.5    10 |   0.0167 /  0.0168   |   0.0935 /  0.0934
  0.5    50 |   0.0008 /  0.0008   |   0.0045 /  0.0046

  -> t0r > t0l at every point (n-side empties later; mu_n=3 mu_p),
     matching the asymmetry of Figs. 5, 8, 9 in the paper.

[ Sections IV-VII: space charge & voltage recovery ]
  boundary speed ratio Rl/Rr = 3.000   (= mu_n/mu_p = 3.0)
  boundaries meet at t/tau   = 0.1120
  left voltage share (equal t0) = 90.0%   (paper: ~90% for b=3)
  p-i-n voltage law  U ~ (t-t0)^1.531   (theory 3/2 = 1.5)
  doped Schottky law U ~ w^1.981          (theory 2)
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Figs. 7 & 8 -- carrier removal (recombination / sweep-out)

MODEL Fig. 8: iR = 10 iF (dots = analytic Eq. 36)



PAPER p.1337: Fig. 7 (iR=0) and Fig. 8 (iR=10iF)

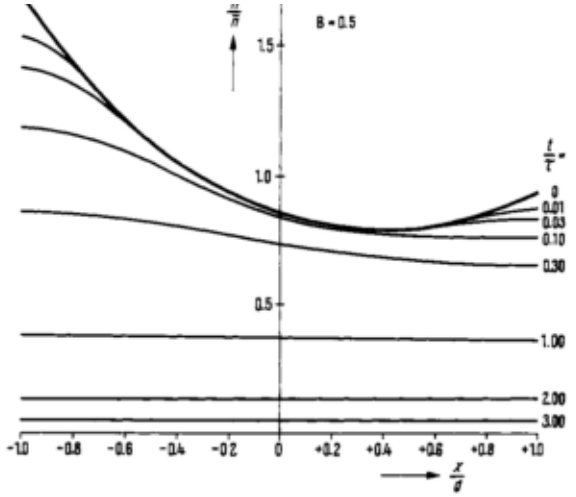


Fig. 7. Removal of the carriers from the swamped zone by recombination with $i_R = 0$. $\mu_n = 3\,\mu_p$; $d = L$.

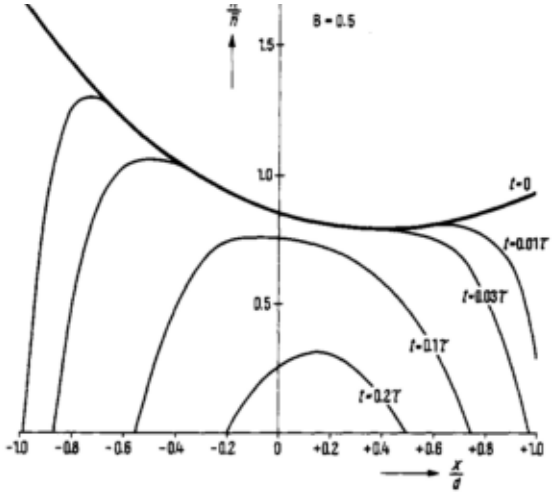


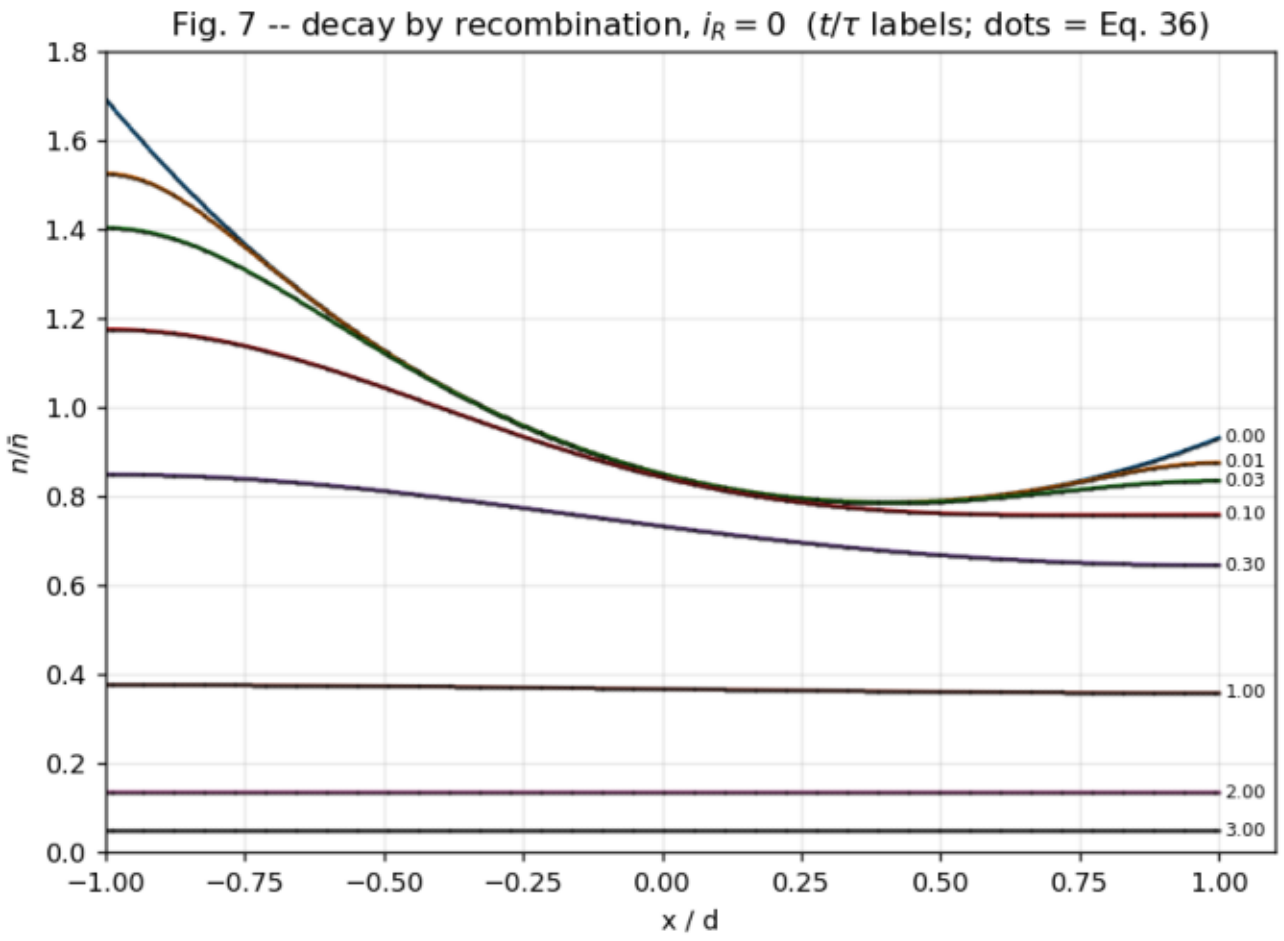
Fig. 8. Sweeping out of the carriers from the swamped zone by a strong reverse current $i_R = 10 i_F$. Beginning of the process. $\mu_n = 3\mu_p$; $d = L$.

begins to flow through the rectifier, the concentration gradient at the right border of the middle region momentarily goes from a positive to a negative value. Likewise, the left

components U_l and U_r of the total rectifier voltage U_{rect}. This is also true for the second field strength component in (31) which depends only on the concentration distribution.

Fig. 7 -- decay by recombination (iR = 0)

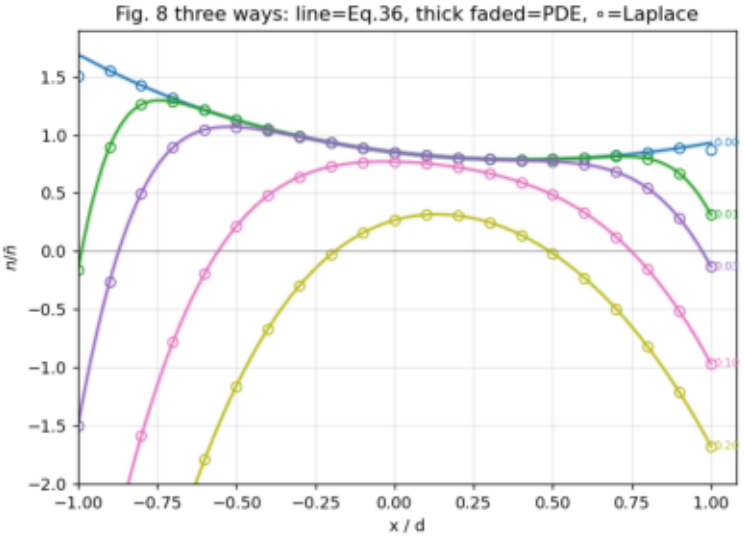
MODEL Fig. 7: t/tau = 0 ... 3.0

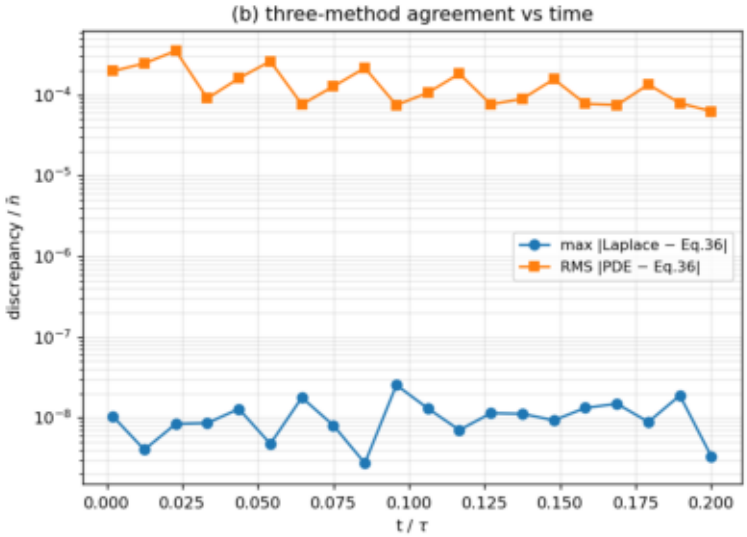


Validation -- sweep-out solved 3 independent ways

PDE / Fourier series Eq.36 / Laplace-Talbot agree: Laplace≡Eq.36 to 1e-8, PDE to 1e-4

Reverse-recovery sweep-out solved 3 independent ways (PDE / Eq.36 series / Laplace-Talbot)





Validation -- solver convergence & cost

Fourier: Gaussian (5 terms->1e-4); Talbot: spectral (6 nodes->1e-4, opt M~16); PDE: O(dt) (1600 steps->1e-4)

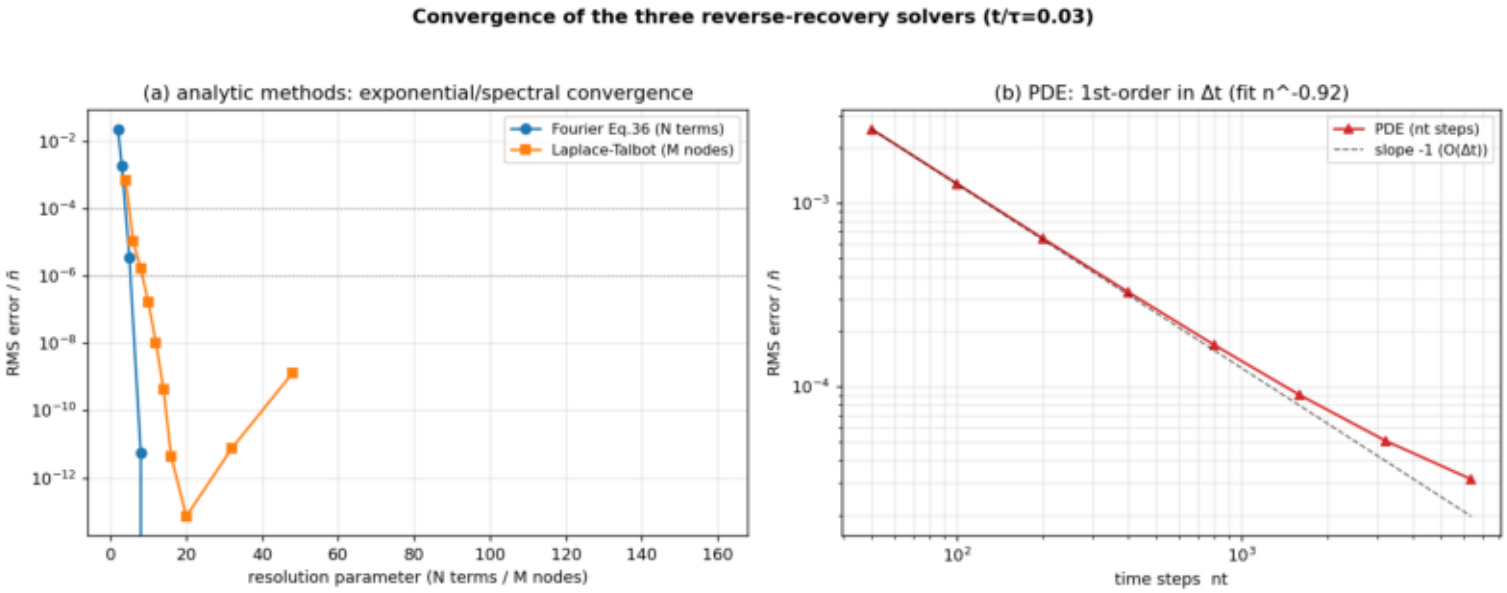
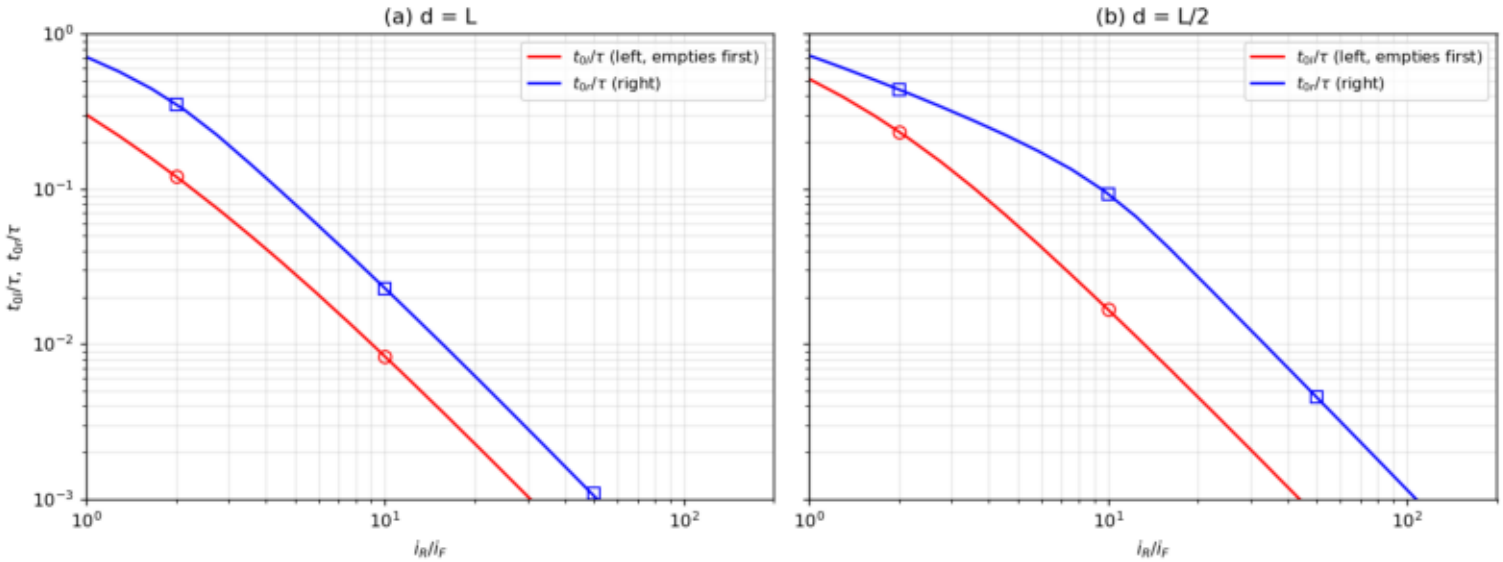


Fig. 9 -- storage (delay) times vs iR/iF

MODEL Fig. 9: t0l/tau, t0r/tau (open markers = numeric PDE)



PAPER p.1338: Fig. 9 (a) d=L (b) d=L/2

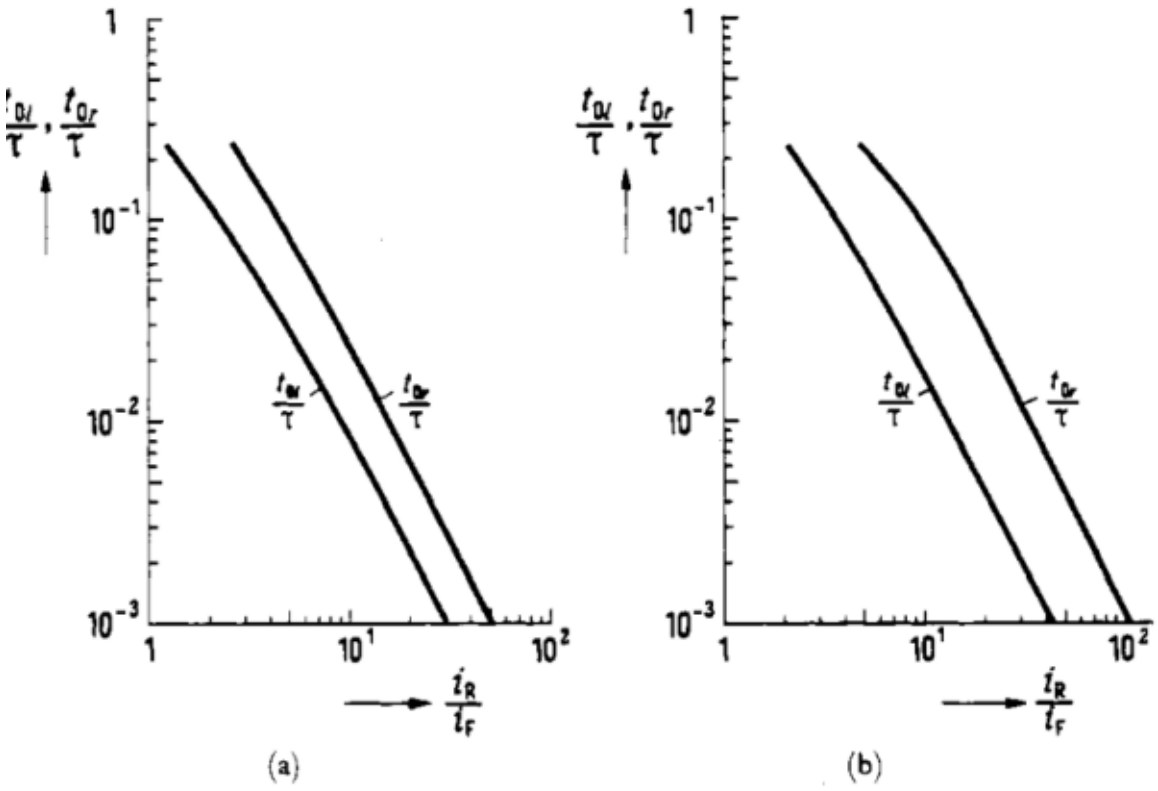


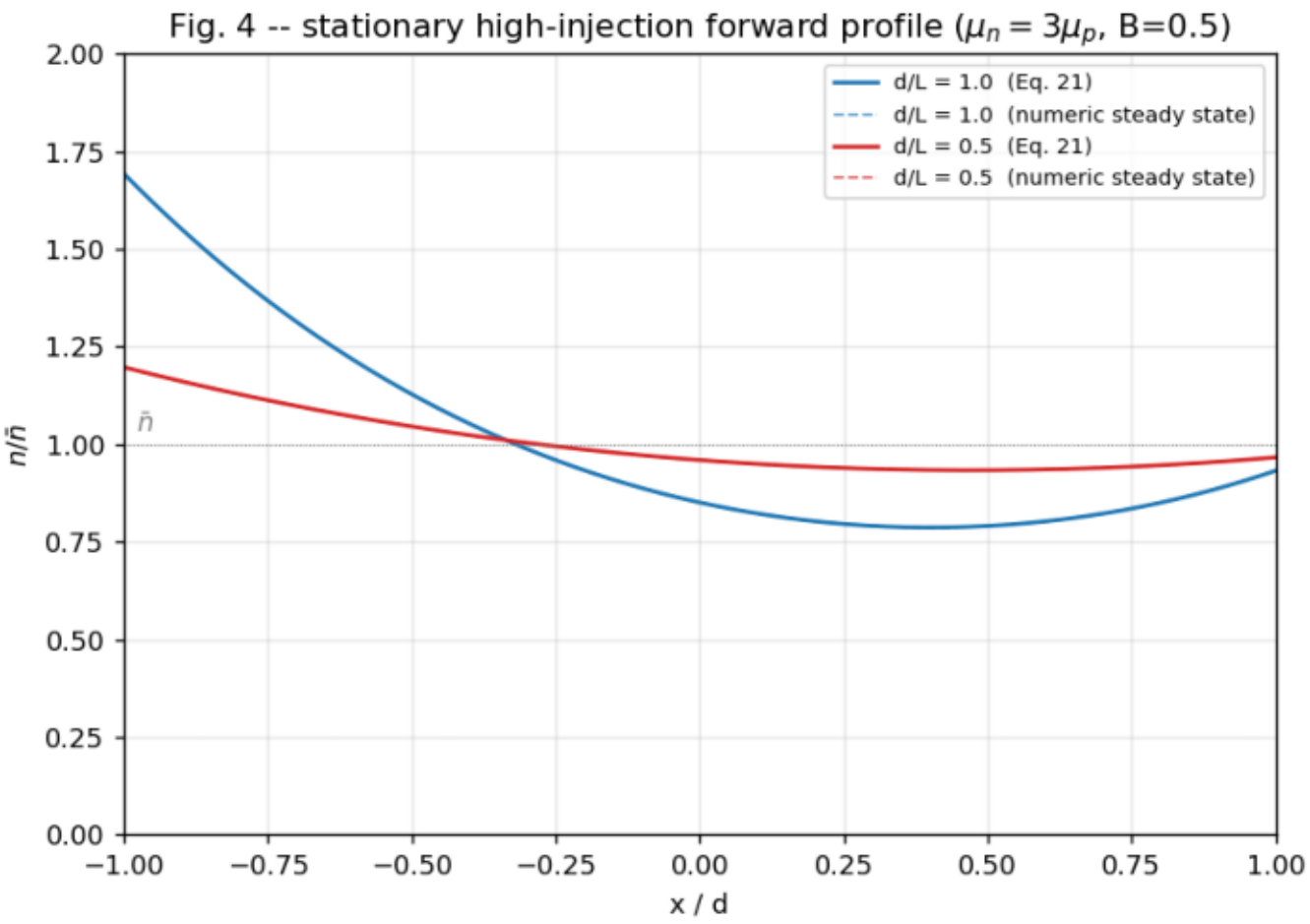
Fig. 9. Dependence of delay times t_{0l} and t_{0r} upon i_R/i_F with negligible doping. (a) $d=L$. (b) $d=L/2$.

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Fig. 4 -- stationary forward profile

MODEL Fig. 4: n/nbar vs x/d (Eq. 21 + numeric check)



Sections IV-VII -- space charge & voltage recovery

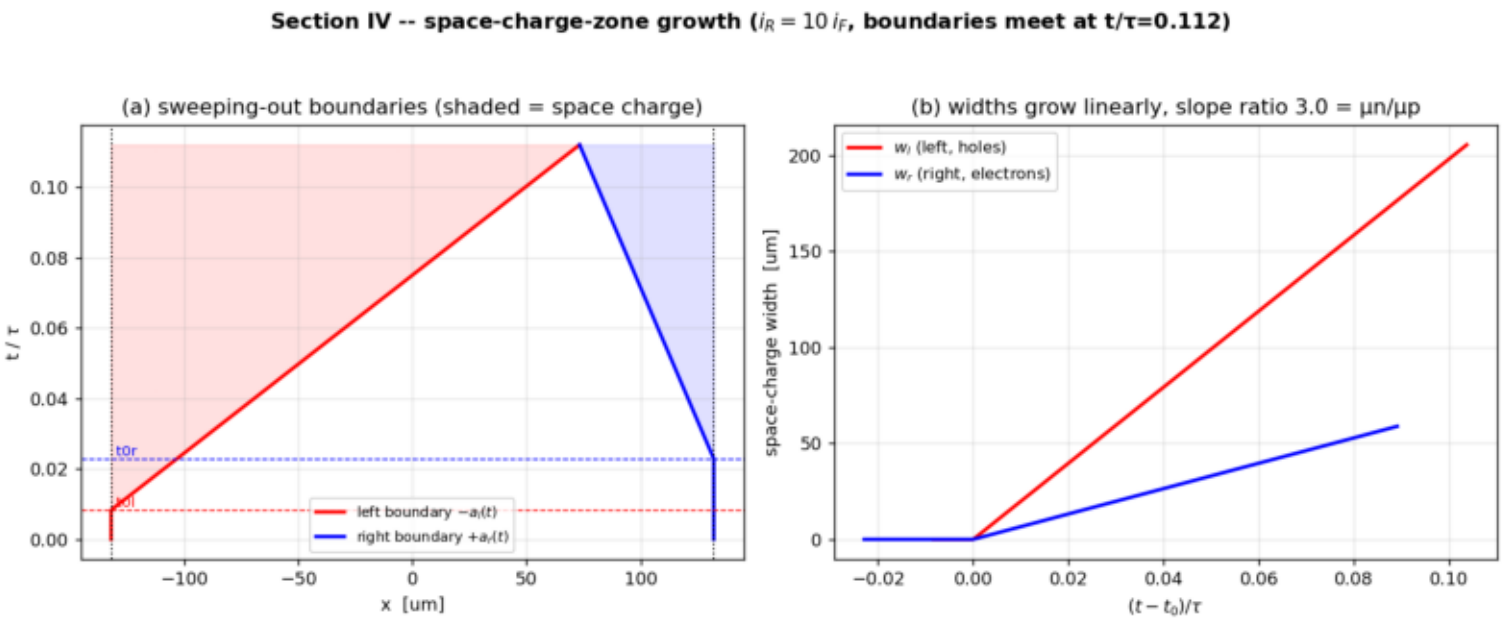
After the storage times t0l, t0r the quasi-neutral plasma model hands off to the space-charge stage:

  Sec. IV two sweeping-out boundaries advance inward; widths
           grow linearly, left mu_n/mu_p = 3x faster (Eq. 46-49).
  Sec. V   p-i-n: sqrt-field in each zone (Eq. 54) -> voltage
           U_sc ~ w^{3/2} ~ (t-t0)^{3/2} (Eq. 58-61); left ~90%.
  Sec. VI uniform doping -> Schottky zone, U_sc ~ w^2.
  Sec. VII diffused (exponential) doping: general doped solver.

Verified: boundary-speed ratio = mu_n/mu_p exactly; left voltage share = 90.0% for b=3 (paper's number); fitted voltage exponents 1.53 (p-i-n, theory 1.5) and 2.02 (Schottky, theory 2).

Sec. IV -- space-charge-zone growth

MODEL: boundary trajectories & linear width growth



PAPER Figs. 11-13 (schematic sweeping-out)

Fig. 10. Distribution of the charge carriers in a *p-i-n* rectifier during the sweeping-out process (schematically).



Fig. 11. Growing of the space-charge zones during the sweeping-out process (schematically).

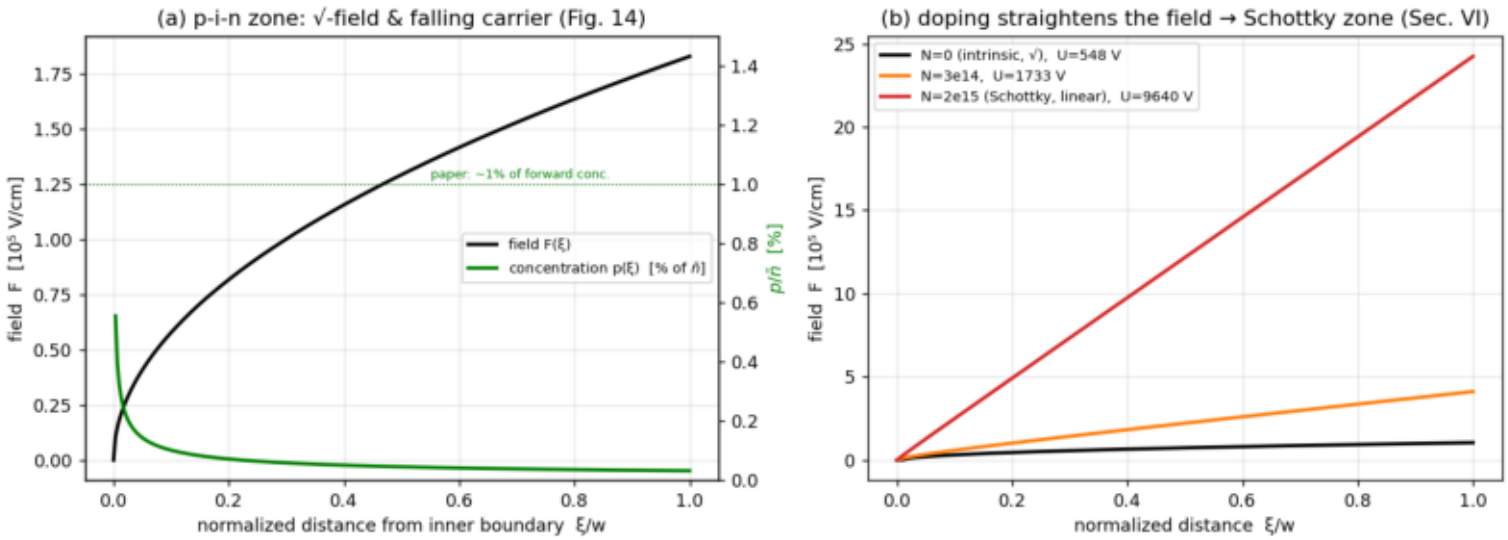
$$i_R \gg i_F. \tag{41}$$

Since the forward current density i_F replaces the stored charge exactly during one lifetime τ in accordance with (24), a much larger sweeping-out current density must remove the stored charge much faster than within a lifetime. Thus, the recombination cannot become effective at all; on the

Sec. V/VI -- space-charge field & carriers

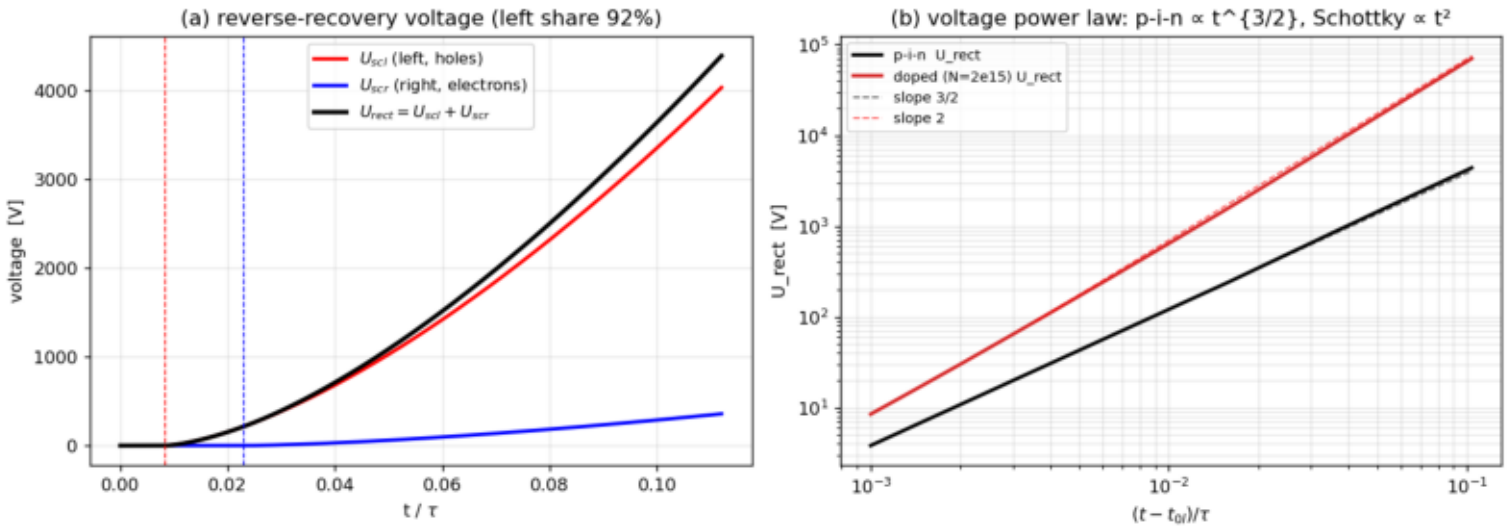
MODEL: sqrt-field + falling carrier (Fig. 14); doping -> linear field (Fig. 16)

Sections V-VI -- space-charge field / carrier profiles



Sec. V -- reverse-recovery VOLTAGE

MODEL: U_rect(t) ~ (t-t0)^{3/2} (p-i-n) / ^2 (Schottky); left zone ~90%



Extension -- Si vs 4H-SiC vs GaN

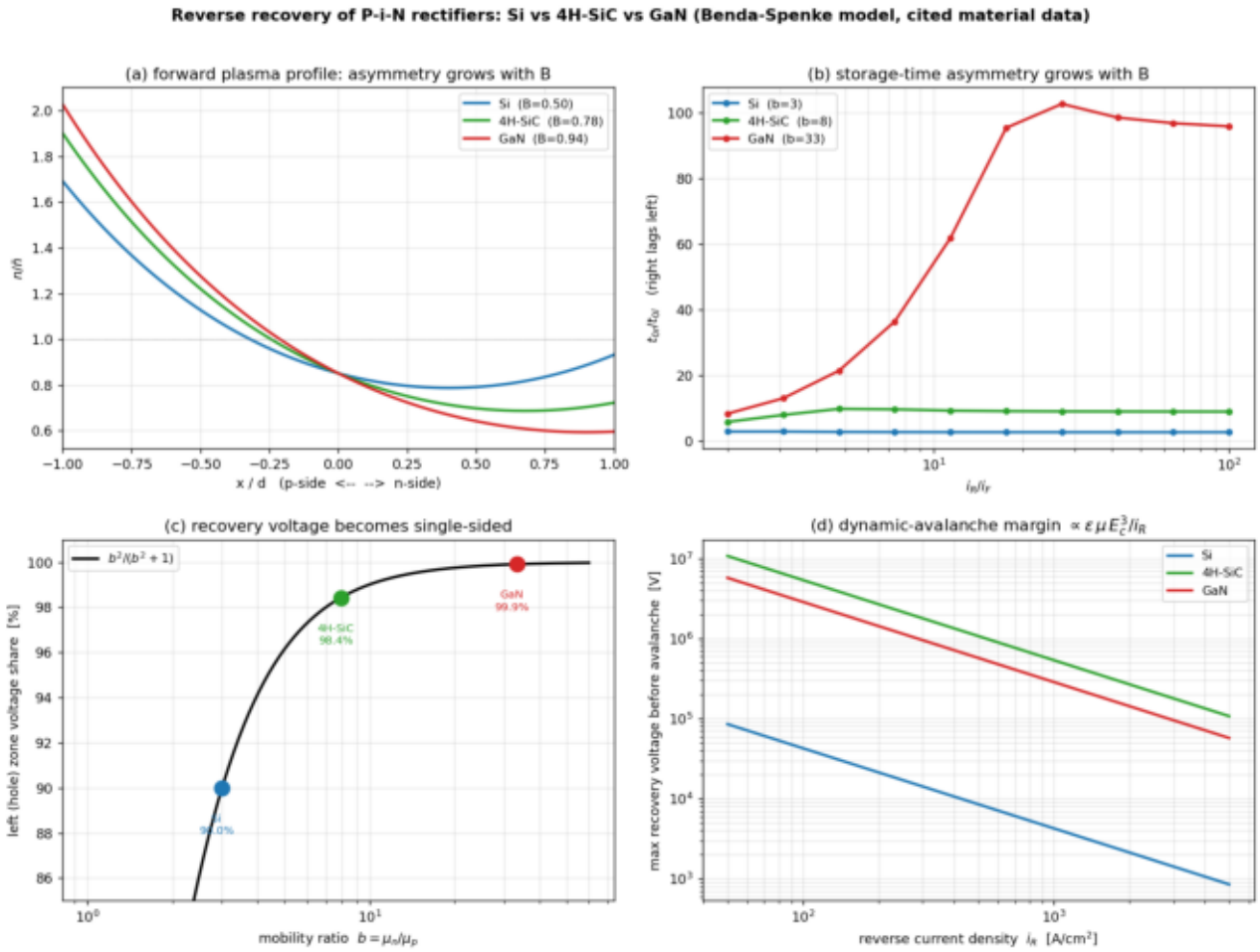
The silicon model is re-run with cited 4H-SiC and GaN parameter sets (materials.py). Going Si -> SiC -> GaN:

* mobility asymmetry B rises 0.50 -> 0.78 -> 0.94, so the forward plasma and the recovery become increasingly one-sided;
* the left (hole) zone's voltage share climbs 90% -> 98% -> ~100%;
* the recovery space charge can hold ~70-130x more voltage before dynamic avalanche -- the Baliga FOM (eps mu Ec^3) emerging from the reverse-recovery physics.

NB GaN's hole mobility (hence B, and its hole-zone avalanche margin) is the most uncertain input; see materials.py.

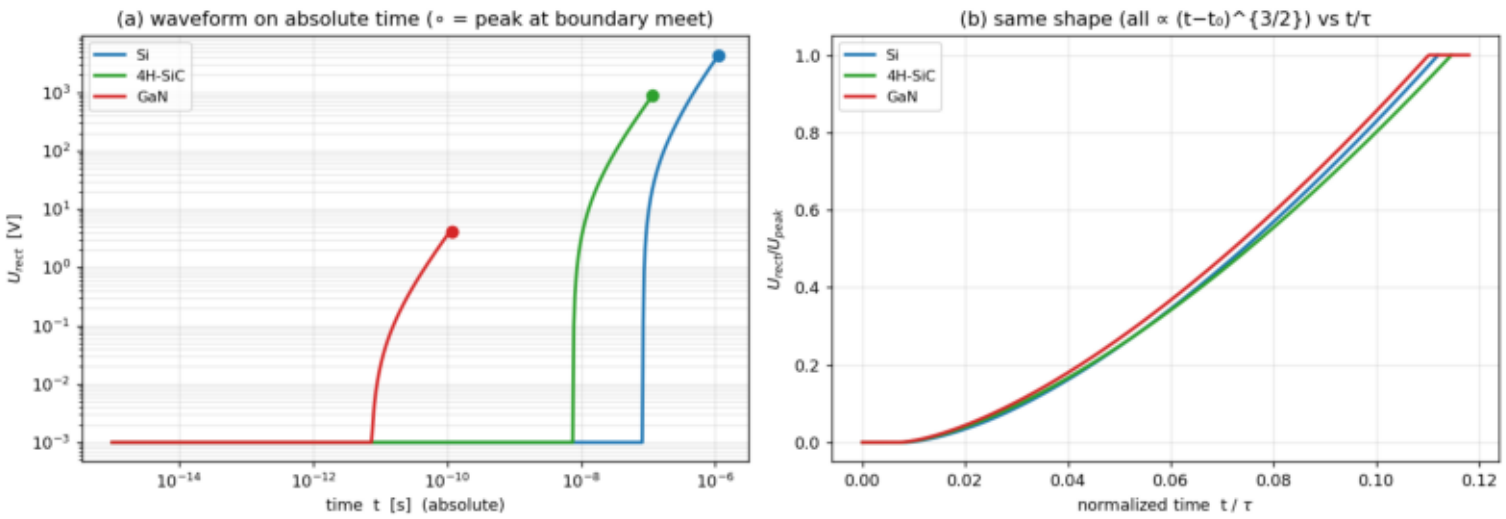
Extension -- reverse recovery: Si / 4H-SiC / GaN

MODEL: (a) forward profile (b) storage-time asymmetry (c) voltage split (d) avalanche margin



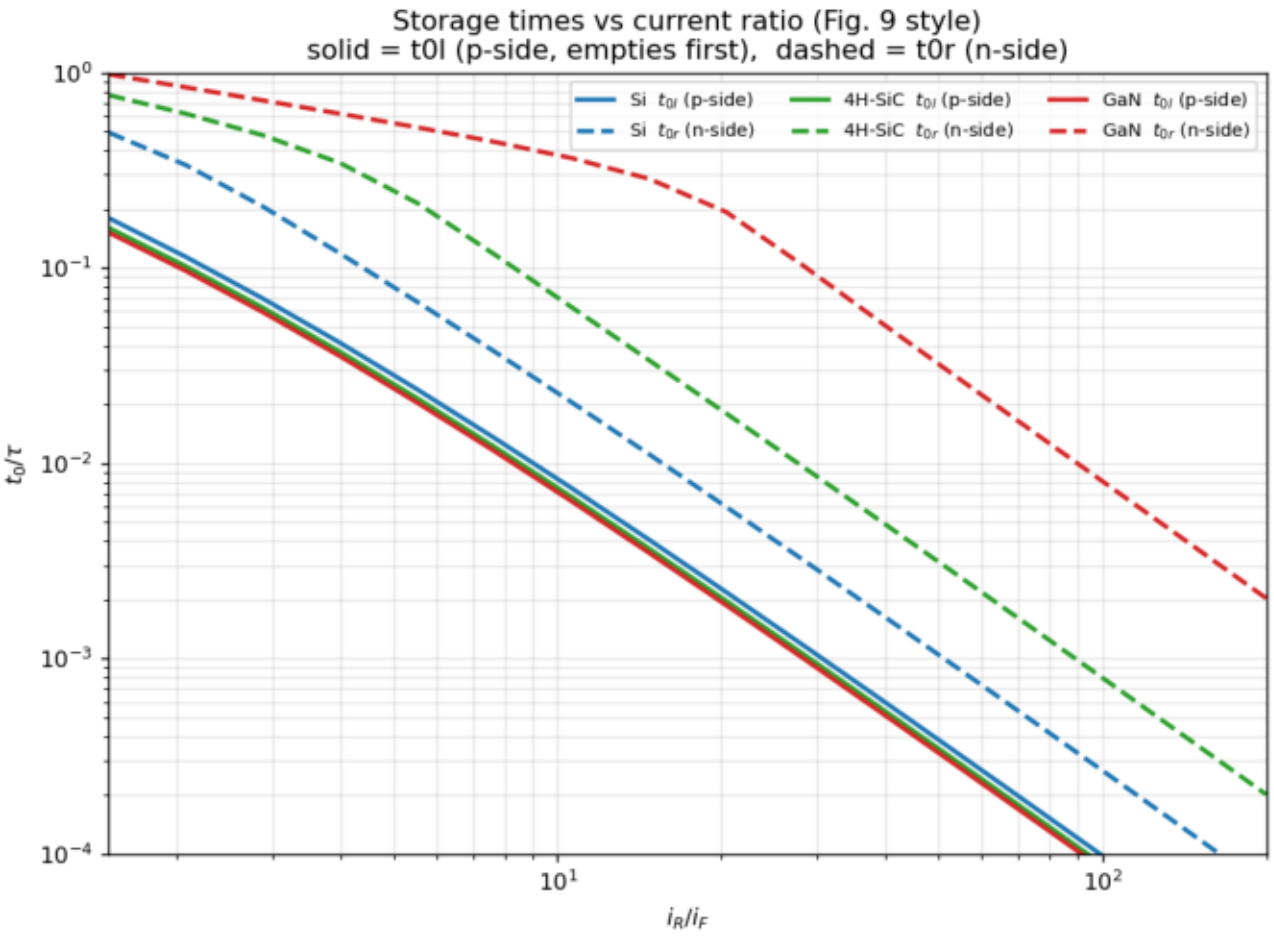
Extension -- fig_mat_waveform.png

MODEL: U_rect(t) waveform, absolute time (WBG recovers ~10^4x faster)



Extension -- fig_mat_fig9.png

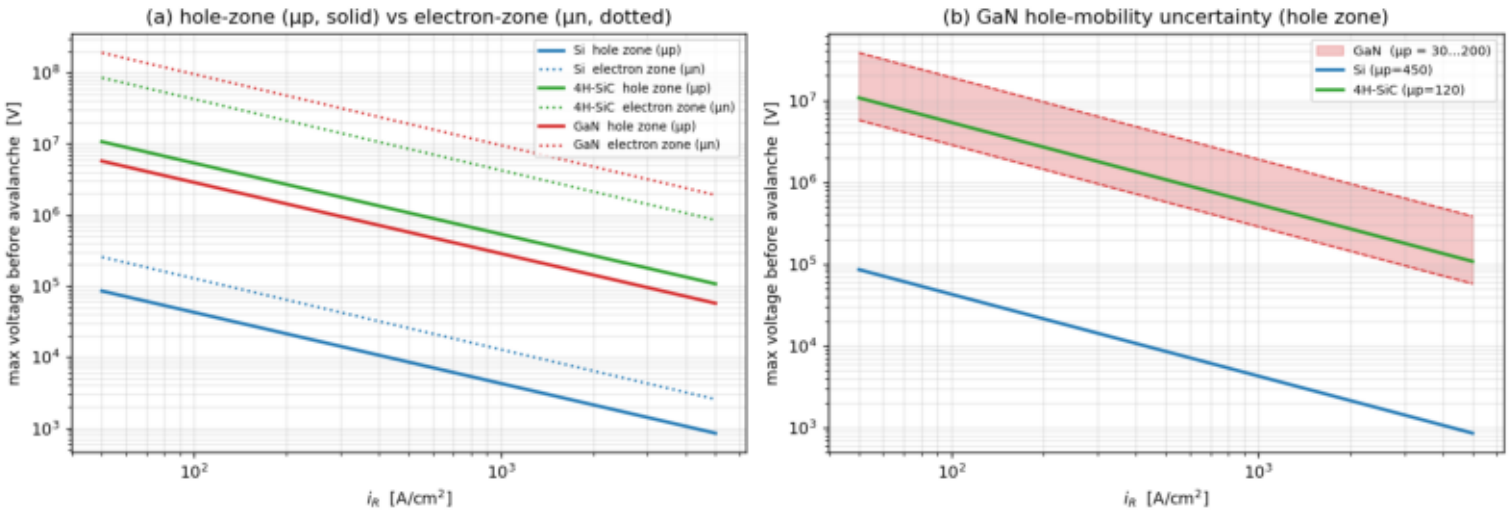
MODEL: storage times t0l/t0r vs iR/iF (Fig. 9 style); n-side fans out



Extension -- fig_mat_avalanche.png

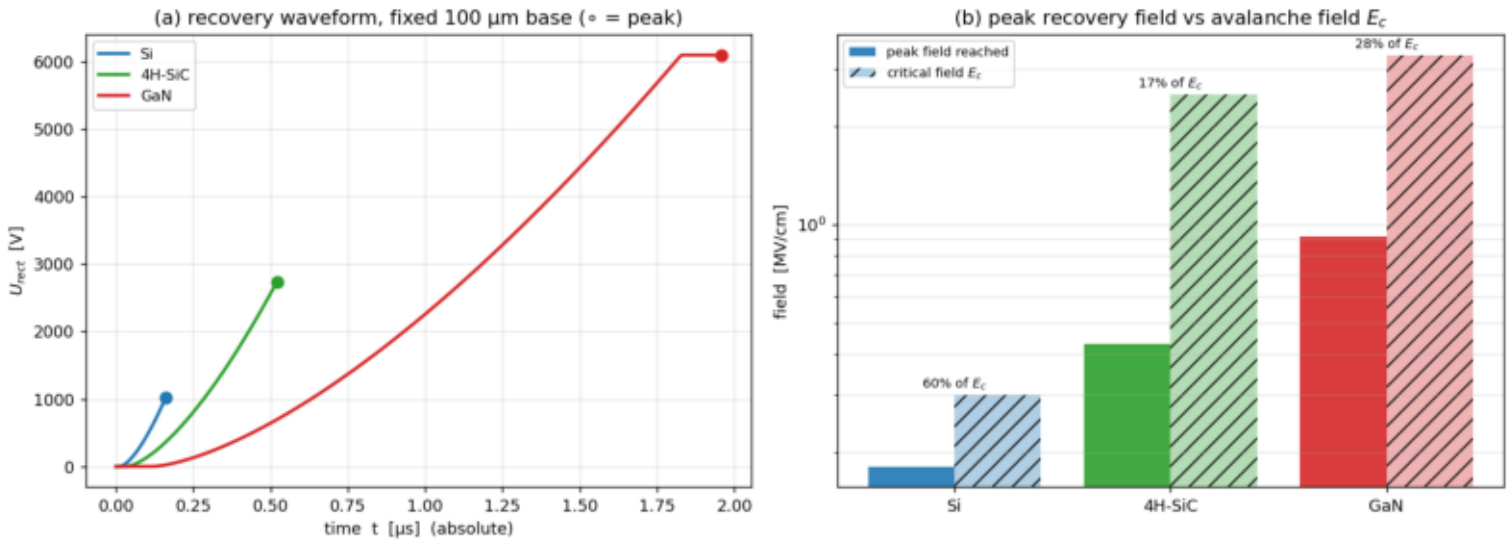
MODEL: avalanche margin: hole vs electron zone, and GaN-mu_p band

Dynamic-avalanche margin ∝ ε·μ·Ec³ / iR (zone-mobility and GaN-μp sensitivity)



Extension -- fig_mat_fixedgeom.png

MODEL: fixed 100um base: comparable peaks; Si reaches 60% of Ec, SiC 17%, GaN 28%



Extension -- conformal mapping: 2-D space-charge edge field

Conformal mapping does NOT apply to the 1-D diffusion transient.
It DOES apply to the doped/Schottky space-charge zone (Sec. VI):
constant charge rho=qN -> phi = phi_p(y) + psi(x,y), a 1-D parabolic
particular part + a SOURCE-FREE harmonic psi (same split as the SOI
Nakagawa Fig-4 method). At a 90-deg junction/contact corner the
map zeta = z^{2/3} gives the classic r^{-1/3} field crowding
|E|/F_1D = (w/r)^{1/3}. Applied to the fixed-100um-base fields:
the 2-D edge crowds the field ~4x, so Si (258% of Ec) and GaN
(129%) reach EDGE avalanche while their 1-D bulk looked safe;
4H-SiC (78%) survives. Edge termination -- invisible to the
paper's 1-D model -- sets reverse-recovery ruggedness.

Harmonicity self-check: max |Laplacian(phi)| ~ 2e-7.

Extension -- conformal edge field crowding

MODEL: (a) corner field map (zeta=z^{2/3}) (b) enhancement + edge avalanche

Conformal-mapping extension: 2-D space-charge edge field (Section VI, doped zone)

